

Dataset Overview

1. Introduction

The dataset used for this project is the **ITSM Service Desk Operations Analytics Dataset**.

It represents an IT Service Management (ITSM) environment in which a managed IT services company supports multiple enterprise clients.

The database is designed to capture:

- Service tickets
- Support users
- SLA tracking
- Operational KPIs
- Management summaries
- Monthly ticket activity

The database is created under the PostgreSQL schema `itsm_ops`.

The SQL script defines the database structure and generates the project data for analysis.

2. Dataset Time Period

The main ticket data is generated for the period:

April 2021 to July 2021

The monthly activity-log tables are also organized for:

- April 2021
- May 2021
- June 2021
- July 2021

Daily KPI data is generated across the same April–July period.

3. Dataset Size

The SQL script is designed to generate:

Data	Records
Company	450
Users	1,250
Tickets	180,000
Ticket SLA records	360,000

The SLA table contains **two SLA records per ticket** because the script creates:

- First Response SLA
- Resolution SLA

Other table sizes are generated from the ticket activity and KPI logic, so their final counts should be validated from the PostgreSQL database rather than assumed.

4. Database Tables

Table	Role / Purpose	Key Relationship
company	Master table containing client/company details.	company_id is the primary key.
app_user	Master table containing support agents and other users.	user_id is the primary key.
ticket	Main transaction table containing service-ticket information.	company_id → company; requester_user_id and assigned_agent_id → app_user

ticket_sla_log	Stores First Response and Resolution SLA measurements.	ticket_id → ticket
daily_itsm_kpi	Stores daily operational KPI measures by company and support team.	company_id → company
management_itsm_summary	Stores monthly management-level KPI summaries.	company_id → company
ticket_activity_log_2021_april	Stores April 2021 ticket activities.	ticket_id → ticket; performed_by_user_id → app_user
ticket_activity_log_2021_may	Stores May 2021 ticket activities.	Same structure as April activity log.
ticket_activity_log_2021_june	Stores June 2021 ticket activities.	Same structure as April activity log.
ticket_activity_log_2021_july	Stores July 2021 ticket activities.	Same structure as April activity log.

5. Important Columns in the Main Tables

company

- company_id: Unique identifier for each company.
- company_name: Name assigned to the client company.
- industry: Industry of the client.
- company_size: Size category such as Small, Mid Market, Enterprise or Strategic.
- city / state / region: Location information.
- contract_type: Type of support contract.
- support_plan: Bronze, Silver, Gold or Platinum.
- active_flag: Indicates whether the company is active.

app_user

- user_id: Unique identifier for a user.
- user_type: Role such as Support Agent, Team Lead, Incident Manager, Problem Manager, Change Manager or End User.
- support_team: Team responsible for support activities.
- skill_level: Junior, Intermediate, Senior, Lead or Manager.
- shift_name: Morning, Evening, Night or General.
- city / region: User location information.
- active_flag: Indicates whether the user is active.

ticket

- ticket_id / ticket_number: Unique ticket identifiers.
- company_id: Links the ticket to a client company.
- requester_user_id: Links the requester to app_user.
- assigned_agent_id: Links the assigned support person to app_user.
- opened_at / resolved_at / closed_at: Important ticket lifecycle timestamps.
- ticket_type: Incident, Service Request, Problem, Change Request or Access Request.
- category / sub_category: Technical classification of the ticket.
- priority: P1, P2, P3 or P4.
- impact / urgency: Business/operational importance indicators.
- ticket_status: Current lifecycle status.
- source_channel: Email, Phone, Portal, Chat, Monitoring Alert or Walk-in.
- reopen_count: Number of times a ticket was reopened.
- customer_satisfaction_rating: Rating from 1 to 5 when available.
- created_month: Month in which the ticket was opened.

ticket_sla_log

This table records two SLA types:

- First Response
- Resolution

It contains:

- SLA target time
- Actual time
- SLA status
- Breach minutes

The possible SLA statuses defined in the database are:

- Met
- Breached
- Running
- Not Applicable

6. KPI and Management Tables

The daily_itsm_kpi table contains daily measures such as:

- Total tickets
- New tickets
- Resolved tickets
- Closed tickets
- Reopened tickets
- Backlog
- SLA met/breached counts
- Average resolution time
- Average first-response time
- Average CSAT

The `management_itsm_summary` table contains monthly management measures including:

- Total tickets
- Incidents
- Service requests
- High-priority tickets
- Resolved/closed tickets
- Backlog
- SLA compliance
- Average resolution hours
- Average CSAT
- Escalation count

7. Analytical Views

The SQL script also creates reporting views.

The main views are:

- `vw_ticket_detail`
- `vw_sla_performance`
- `vw_daily_kpi_summary`
- `vw_management_itsm_dashboard`
- `vw_activity_log_all_months`

These views combine or summarize information for easier reporting.

8. Data Flow for the Project

The project uses PostgreSQL as the database layer and Jupyter Notebook with Python for data preparation and analysis.

The required data is imported into Pandas DataFrames.

The two DataFrames required by the project are then **merged in Jupyter Notebook using Pandas**.

After the merge, the prepared dataset is used for:

1. Data Cleaning & Preparation
2. Univariate Analysis
3. Bivariate Analysis
4. Multivariate Analysis

9. Primary Keys and Relationships

- company.company_id is the primary key of the company table.
- app_user.user_id is the primary key of the app_user table.
- ticket.ticket_id is the primary key of the ticket table.
- ticket.company_id references company.company_id.
- ticket.requester_user_id references app_user.user_id.
- ticket.assigned_agent_id references app_user.user_id.
- ticket_sla_log.ticket_id references ticket.ticket_id.
- daily_itsm_kpi.company_id references company.company_id.
- management_itsm_summary.company_id references company.company_id.
- Each monthly activity log links back to ticket through ticket_id and can link to app_user through performed_by_user_id.

10. Dataset Understanding Before Analysis

Before performing EDA, the dataset should be checked in Python for:

- Row counts
- Column names
- Data types
- Missing values
- Duplicate records

- Logical consistency

THANKYOU!!!!
